# Transfer: circuits and induction

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Apply circuit relationships and explain an induction observation.

**SUCCESS CRITERIA - I can...**

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A device receives **50 J** mechanical energy at its shaft and produces electrical energy at its terminals. **Name it** and **name one possible loss**.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $V = IR$; $P = IV$; $E = P\Delta t$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$. Use stated facts about the circuit or coil. Ideal battery and wires unless stated otherwise.

**Worked reasoning:** A **12 V** supply drives a **6 ohm** resistor: current **2 A** and power **24 W**. A complete explanation checks both the charge-flow rate and the energy-transfer rate.

**Your turn:** 1. **Draw the circuit** and an energy-flow arrow.
2. **Label what an ammeter** and voltmeter would measure and where each connects.




Label component values, current and voltage as needed.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

**Sketch two resistors** in parallel across a battery and **label the equal-voltage branches**.




Circuit symbols: long/short lines = battery; rectangles = resistors.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A **6 V** battery powers **3 Ω** and **6 Ω** resistors in parallel.
1. **Label both branch voltages**.
2. **Find each current**, total source current and total power $P = VI_{total}$.




Label component values, current and voltage as needed.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it. **Explain the change** in the meter response without saying the magnet supplies unlimited energy.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A **4 Ω** resistor carries **6 A**. **Find the potential difference across** it.
**Draw the stated circuit** and **label quantities relevant** to the question.

Draw the complete circuit and label shared nodes.

**Marzano check - circle:** 1 2 3

## 07 Exit check

A motor uses **6 W** for **5 s**. **Find energy** and **name one possible non-mechanical output**.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

Replace the resistor with two **12 ohm** resistors in parallel. **Compare total current** and power, then **draw both branch currents**.




Label component values, current and voltage as needed.

**Marzano check - circle:** 1 2 3